a

Task

Colour fundus photograph in, four lesion probability maps out



- **MA** microaneurysm
- **HE** hemorrhage
- **EX** hard exudate
- **SE** soft exudate

Data collection

8 test sets, 7 datasets · 5,121 images

Training sources	
IDRiD	81
DDR	757
Independent sources	
TJDR-std	257
TJDR-uwf	304
MAPLES-DR	198
FGADR	1,842
Retinal-Lesions	1,593
DiaRetDB1	89

Model collection

39 variants in the registry

- Tier 1 · main table — **22**
- Tier 2 · if time allows — **6**
- Tier 3 · paper audit only — **11**

Reproducibility level R0–R4 recorded for every variant; failures stay in the denominator

Four evaluations

- **Paper audit** — numbers as reported, and where they came from
- **Released checkpoints** — scored with the paper's number, the authors' metric and ours
- **Author-recipe reruns** — retrained with the authors' own pipeline
- **Unified retraining** — shared data, budget, augmentation, selection

Tests and metrics

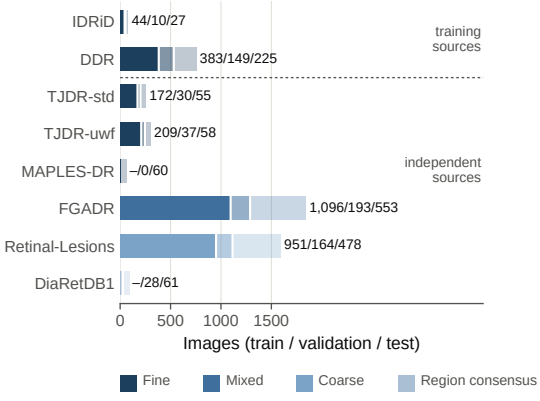
- **Internal test** — official test set of the training source
- **Independent test** — sources never used for training

Primary — exact pooled AUPR inside the field of view; mAUPR

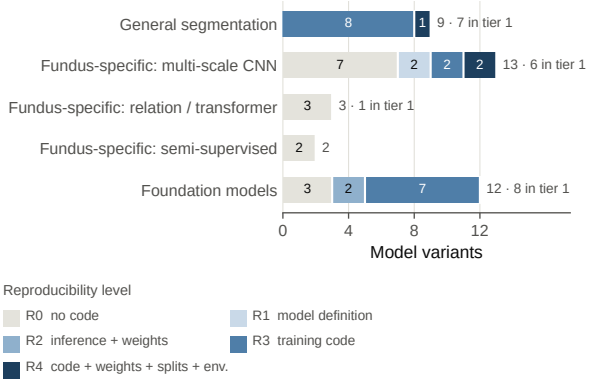
Secondary — Dice, IoU; lesion-level recall by lesion size

Statistics — 3 seeds; image bootstrap; Friedman + Nemenyi

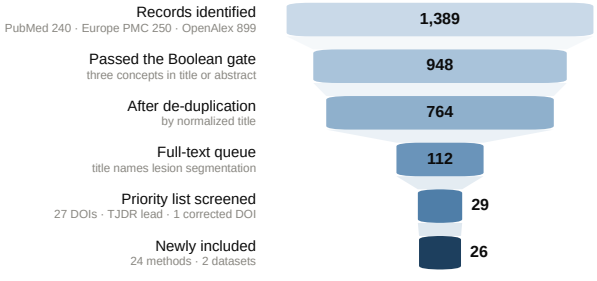
b Test sources and frozen splits



c Registered models by family and reproducibility level



d Literature search and screening



Fixed search 2015-01-01 to 2026-09-30; with the snowballed corpus the ledger holds 86 records.

e Factors assessed

Training source	Label granularity	Training amount	Input resolution
5 representative models	*U-Net · M2MRF*	*U-Net · M2MRF*	*M2MRF · RETFound*
IDRiD · DDR · TJDR-std · FGADR	fine · mixed · coarse, alone and combined	10 · 50 · 100% of DDR; fine sources merged	512 · 1,440 canvas · native-resolution patches

Loss	Learning rate	Foundation models	Encoder pretraining
U-Net	*HRNet-W48 · M2MRF*	*8 models*	*4 encoders*
CE · Dice · CE + Dice · CE + per-image Dice	recipe value × ⅓, 1, 3; validation only	zero-shot · oracle prompt PEFT · encoder + decoder	retinal vs natural-image ViT-L, one UPerNet head

One factor changes at a time; everything else stays fixed. The five representative models are U-Net, nnU-Net, M2MRF, SAM2-UNet and RETFound + UPerNet.